PADHAKU
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CHAPTER 1

Orienting Yourself: The Use of Coordinates

NCERT Exercise Solutions

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✨ Try each question yourself first, then check the answer. 💡
answers are samples: write your own.

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⭐ 1.2 Settling In

? Fig. 1.1 shows only the map of the floor. Why can the position of the windows not be marked on this map?

✔ ANSWER

The map shows only the floor, which is flat (2-D: length and width).

Windows are on the walls, at some height above the floor.

A floor map has no way to show height, so **windows cannot be marked on it**.

📝 Exercise Set 1.1

(i) If D_1R_1 represents the door to Reiaan's room, how far is the door from the left wall (the y-axis)? How far is the door from the x-axis?

✔ ANSWER

In Fig. 1.3, D_1 is at the mark 8 on the x-axis.

Distance from the left wall (y-axis) = **8 ft**.

The door lies on the x-axis, so its distance from the x-axis = **0 ft**.

(ii) What are the coordinates of D_1?

✔ ANSWER

D_1 = (8, 0)

(iii) **If R_1 is the point (11.5, 0), how wide is the door? Is this a comfortable width? Can a person in a wheelchair enter easily?**

✔ ANSWER

Width = D_1R_1 = 11.5 − 8 = **3.5 ft** (about 107 cm).

This is a comfortable width for a room door.

A wheelchair needs a door about 3 ft (90 cm) wide, so **yes, a wheelchair user can enter easily**.

(iv) **If B_1 (0, 1.5) and B_2 (0, 4) are the ends of the bathroom door, is the bathroom door narrower or wider than the room door?**

✔ ANSWER

Bathroom door = B_1B_2 = 4 − 1.5 = 2.5 ft.

Room door = 3.5 ft.

2.5 < 3.5, so the bathroom door is **narrower** (by 1 ft).

🤔 Think and Reflect · Exercise Set 1.1

1 **What are the standard widths for a room door? Look around your home and in school.**

💡 SAMPLE ANSWER

Most room doors are about **3 ft (about 90 cm)** wide; bathroom doors are often narrower, about 2.5 ft (75 cm).

Measure the doors at home and in school with a tape and compare.

2 **Are the doors in your school suitable for people in wheelchairs?**

💡 SAMPLE ANSWER

A wheelchair needs a clear door width of about 3 ft (90 cm), no step (or a ramp) and space to turn.

Example: our classroom doors are about 3 ft wide, so they are suitable, but the toilet door is narrower and has a step, so it is not.

⭐ 1.3 Quadrants (in-text)

? **Copy Fig. 1.4 and mark S (3, −5) and Q (−5, 3). Mark any point P in Quadrant I and any point R in Quadrant III, and write their coordinates.**

💡 SAMPLE ANSWER

S (3, −5): 3 units right of O, then 5 units down → Quadrant IV.

Q (−5, 3): 5 units left of O, then 3 units up → Quadrant II.

Example: **P (2, 3)** in Quadrant I (both coordinates positive).

Example: **R (−4, −2)** in Quadrant III (both coordinates negative).

🤔 Think and Reflect · 1.3 Quadrants

1 **What is the x-coordinate of a point on the y-axis?**

✔ ANSWER

A point on the y-axis is at distance 0 from the y-axis.

So its x-coordinate is **0**; the point is (0, y).

2 **Is there a similar generalisation for a point on the x-axis?**

✔ ANSWER

Yes. A point on the x-axis has y-coordinate **0**; the point is (x, 0).

3 **Does point Q (y, x) ever coincide with point P (x, y)? Justify your answer.**

✔ ANSWER

Two points are the same only if their x-coordinates are equal and their y-coordinates are equal.

(y, x) = (x, y) means y = x.

So Q coincides with P **only when x = y**, e.g. (4, 4). If x ≠ y, e.g. (2, 5) and (5, 2), they are different points.

4 **If $x \neq y$, then $(x, y) \neq (y, x)$; and $(x, y) = (y, x)$ if and only if $x = y$. Is this claim true?**

✔ ANSWER

Yes, the claim is true.

If $x = y$, both pairs are the same, so $(x, y) = (y, x)$.

If $x \neq y$, the first coordinates x and y are different, so $(x, y) \neq (y, x)$.

📝 Exercise Set 1.2

1 (i) **Reiaan's rectangular study table has three feet at (8, 9), (11, 9) and (11, 7). Where will the fourth foot be?**

✔ ANSWER

$(8, 9)$ and $(11, 9)$ are on the line $y = 9$; $(11, 9)$ and $(11, 7)$ are on the line $x = 11$.

The fourth foot must be below $(8, 9)$ and to the left of $(11, 7)$: $x = 8$, $y = 7$.

Fourth foot = **$(8, 7)$**.

1 (ii) **Is this a good spot for the table?**

💡 SAMPLE ANSWER

Yes. The table (x from 8 to 11, y from 7 to 9) is near the right wall.

It does not touch the bed (x up to about 6.5), the wardrobe (y up to 2) or the door (on the x-axis), and leaves the way from the door free, which is important for Reiaan.

1 (iii) **What is the width of the table? The length? Can you make out the height of the table?**

✔ ANSWER

Length = 11 − 8 = **3 ft**.

Width = 9 − 7 = **2 ft**.

No, the height cannot be found: the map shows only the floor (2-D), not heights.

2 **If the bathroom door has a hinge at B_1 and opens into the bedroom, will it hit the wardrobe? What changes would you suggest if the door is made wider?**

✔ ANSWER

Door width = 2.5 ft, so while opening, its free end moves at most 2.5 ft from B_1 (0, 1.5).

The wardrobe begins at x = 3 (W_1 = (3, 0), W_4 = (3, 2)), i.e. 3 ft from the y-axis.

2.5 < 3, so **the door will not hit the wardrobe** (gap 0.5 ft).

If the door is made wider than 3 ft, it will hit the wardrobe. Then: hinge the door at B_2 (top end), make it open into the bathroom, use a sliding door, or move the wardrobe to the right.

3 (i) **What are the coordinates of the four corners O, F, R and P of the bathroom?**

✔ ANSWER

The bathroom is 6 ft × 9 ft, to the left of the y-axis.

O (0, 0), F (0, 9), R (−6, 9), P (−6, 0)

3 (ii) **What is the shape of the showering area SHWR? Write the coordinates of the four corners.**

💡 SAMPLE ANSWER

From Fig. 1.5: **S (−6, 6), H (−3, 6), W (−2, 9), R (−6, 9)**.

SH (on y = 6) and RW (on y = 9) are parallel; SH = 3 ft, RW = 4 ft are unequal, and HW is slanted.

So SHWR is a **trapezium** (RS is perpendicular to both parallel sides, so it is a right trapezium).

3 (iii) **Mark off a 3 ft × 2 ft space for the washbasin and a 2 ft × 3 ft space for the toilet. Write the coordinates of the corners.**

💡 SAMPLE ANSWER

One possible plan (along the left wall, as in Fig. 1.5):

- Washbasin (3 ft along x, 2 ft along y): **(−6, 0), (−3, 0), (−3, 2), (−6, 2)**

- Toilet (2 ft along x, 3 ft along y): **(−6, 3), (−4, 3), (−4, 6), (−6, 6)**

Both fit inside the bathroom and do not overlap the showering area (y ≥ 6).

4 (i) **The dining room is 18 ft long and 15 ft wide; its length extends from P to A. Sketch it and mark the coordinates of its corners.**

✔ ANSWER

P = (−6, 0), A = (12, 0); PA = 12 − (−6) = 18 ft = length. ✓

Reiaan's door is on the x-axis and leads from the dining room, so the dining room is below the x-axis, 15 ft deep.

Corners: **P (−6, 0), A (12, 0), (12, −15), (−6, −15)**.

4 (ii) **Place a 5 ft × 3 ft dining table precisely in the centre of the dining room. Write the coordinates of the feet.**

✔ ANSWER

Centre of the room: x = (−6 + 12)/2 = 3, y = (0 + (−15))/2 = −7.5 → (3, −7.5).

Length 5 ft along x: x from 3 − 2.5 = 0.5 to 3 + 2.5 = 5.5.

Width 3 ft along y: y from −7.5 − 1.5 = −9 to −7.5 + 1.5 = −6.

Feet: **(0.5, −6), (5.5, −6), (5.5, −9), (0.5, −9)**.

⭐ 1.4 Distance Between Two Points (in-text)

? **Triangle ADM has A (3, 4), D (7, 1), M (9, 6). How do we find the lengths of its sides AD, DM and MA?**

✔ ANSWER

Find the shift along x and along y, then use the Baudhāyana–Pythagoras theorem.

AD = $\sqrt{(4^2 + 3^2)} = \sqrt{25} =$ **5 units**

DM = $\sqrt{((9 - 7)^2 + (6 - 1)^2)} = \sqrt{(4 + 25)} =$ **$\sqrt{29}$ units**

MA = $\sqrt{((9 - 3)^2 + (6 - 4)^2)} = \sqrt{(36 + 4)} =$ **$\sqrt{40}$ units**

🤔 Think and Reflect · 1.4 Distance

1 **In moving from A (3, 4) to D (7, 1), what distance is covered along the x-axis? Along the y-axis?**

✔ ANSWER

Along the x-axis: 7 − 3 = **4 units**.

Along the y-axis: 4 − 1 = **3 units**.

2 **Can these distances help you find the distance AD?**

✔ ANSWER

Yes. They are the two perpendicular sides of a right triangle with hypotenuse AD.

AD = $\sqrt{(4^2 + 3^2)} = \sqrt{(16 + 9)} = \sqrt{25} = $ **5 units**.

? **Triangle AMD is reflected in the y-axis (Fig. 1.9). What are the coordinates of the images of A, M and D?**

✔ ANSWER

Reflection in the y-axis changes the sign of the x-coordinate only.

A' (−3, 4), M' (−9, 6), D' (−7, 1)

🤔 Think and Reflect · Reflection

1 **What has remained the same and what has changed with this reflection?**

✔ ANSWER

Same: the side lengths (5, √29, √40), the angles, the shape and size of the triangle, and the y-coordinates.

Changed: the x-coordinates changed sign, the triangle moved from Quadrant I to Quadrant II, and it is turned the other way (mirror image).

2 **Would these observations be the same if ΔADM is reflected in the x-axis (instead of the y-axis)?**

✔ ANSWER

Yes, lengths, angles, shape and size stay the same.

Now the y-coordinates change sign and x-coordinates stay: A'' (3, −4), D'' (7, −1), M'' (9, −6).

The triangle moves to Quadrant IV.

📝 End-of-Chapter Exercises

1 **What are the x-coordinate and y-coordinate of the point of intersection of the two axes?**

✔ ANSWER

The axes meet at the origin O.

x-coordinate = **0**, y-coordinate = **0**, i.e. O (0, 0).

2 **Point W has x-coordinate −5. Predict the coordinates of point H on the line through W parallel to the y-axis. Which quadrants can H lie in?**

✔ ANSWER

Every point on a line parallel to the y-axis has the same x-coordinate.

So **H = (−5, y)** for some number y.

If y > 0, H is in **Quadrant II**; if y < 0, H is in **Quadrant III**; if y = 0, H = (−5, 0) is on the x-axis.

3 **R (3, 0), A (0, −2), M (−5, −2), P (−5, 2) are joined in order. Predict: (i) two perpendicular sides of RAMP; (ii) a side parallel to one of the axes; (iii) two points that are mirror images in one axis, and which axis.**

✔ ANSWER

AM: A and M both have y = −2, so AM is horizontal.

MP: M and P both have x = −5, so MP is vertical.

(i) **AM ⊥ MP** (a horizontal and a vertical line meet at a right angle).

(ii) **AM is parallel to the x-axis** (also MP is parallel to the y-axis).

(iii) **M (−5, −2) and P (−5, 2)** are mirror images in the **x-axis** (same x, y-coordinates opposite).

Plotting the points confirms these.

4 **Plot Z (5, −6). Construct a right-angled triangle IZN and find the lengths of the three sides.**

💡 SAMPLE ANSWER

One choice: N = (0, 0) (origin), I = (5, 0), Z = (5, −6). Right angle at I.

NI = 5 − 0 = **5 units** (along the x-axis)

IZ = 0 − (−6) = **6 units** (vertical)

NZ = $\sqrt{(5^2 + 6^2)}$ = $\sqrt{(25 + 36)}$ = **$\sqrt{61} \approx 7.81$ units**

(Answers differ with your choice of I and N.)

5 **What would a system of coordinates be like if we did not have negative numbers? Would it allow us to locate all points on a 2-D plane?**

✔ ANSWER

Without negative numbers we could only measure to the right of and above the origin.

We would get only Quadrant I and the positive parts of the axes.

Points to the left of or below the origin (Quadrants II, III, IV) could not be described, so **no, we could not locate all points** of the plane.

***6** **Are M (−3, −4), A (0, 0) and G (6, 8) on the same straight line? Suggest a method to check this without plotting.**

✔ ANSWER

Method: find the three distances. If the sum of the two smaller ones equals the largest, the points lie on one line.

MA = $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$

AG = $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$

MG = $\sqrt{(9^2 + 12^2)} = \sqrt{225} = 15$

MA + AG = 5 + 10 = 15 = MG, so **yes, M, A, G are on the same straight line** (A lies between M and G).

***7** **Use your method to check if R (−5, −1), B (−2, −5) and C (4, −12) are on the same straight line.**

✔ **ANSWER**

RB = $\sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$

BC = $\sqrt{(6^2 + 7^2)} = \sqrt{85} \approx 9.220$

RC = $\sqrt{(9^2 + 11^2)} = \sqrt{202} \approx 14.213$

RB + BC ≈ 14.220, which is not equal to RC ≈ 14.213.

Check without decimals: $(5 + \sqrt{85})^2 = 110 + 10\sqrt{85} \neq 202$, because $10\sqrt{85} \neq 92$ ($\sqrt{85}$ is not 9.2).

Also the steps differ: R→B is 3 right, 4 down; B→C is 6 right, 7 down (not double of 3, 4).

So **R, B, C are not on the same straight line** (they are very close to it, so the plot looks almost straight).

***8** **Using the origin as one vertex, plot the vertices of: (i) a right-angled isosceles triangle; (ii) an isosceles triangle with one vertex in Quadrant III and the other in Quadrant IV.**

💡 **SAMPLE ANSWER**

(i) Example: **O (0, 0), A (4, 0), B (0, 4)**. OA = OB = 4 and the angle at O is 90°.

(ii) Example: **O (0, 0), P (−3, −4), Q (3, −4)**. P is in Quadrant III, Q in Quadrant IV.

OP = √(9 + 16) = 5 and OQ = √(9 + 16) = 5, so OP = OQ: isosceles.

***9** **Is M the midpoint of segment ST? Justify. Then find a connection between the coordinates of M, S and T.**

S	M	T
(−3, 0)	(0, 0)	(3, 0)
(2, 3)	(3, 4)	(4, 5)
(0, 0)	(0, 5)	(0, −10)
(−8, 7)	(0, −2)	(6, −3)

✔ ANSWER

Midpoint of ST = ((x of S + x of T)/2, (y of S + y of T)/2).

S, M, T	Midpoint?	Reason
(−3, 0), (0, 0), (3, 0)	**Yes**	((−3+3)/2, (0+0)/2) = (0, 0) = M; SM = MT = 3
(2, 3), (3, 4), (4, 5)	**Yes**	((2+4)/2, (3+5)/2) = (3, 4) = M; SM = MT = √2
(0, 0), (0, 5), (0, −10)	**No**	SM = 5, MT = 15; midpoint is (0, −5)
(−8, 7), (0, −2), (6, −3)	**No**	SM = √145, MT = √37; midpoint is (−1, 2)

Connection: if S = (x_1, y_1) and T = (x_2, y_2), then **M = $((x_1 + x_2)/2, (y_1 + y_2)/2)$**.

***10** **Find the coordinates of B, given that M (−7, 1) is the midpoint of A (3, −4) and B (x, y).**

✔ ANSWER

$(3 + x)/2 = -7 \rightarrow 3 + x = -14 \rightarrow x = -17$

$(-4 + y)/2 = 1 \rightarrow -4 + y = 2 \rightarrow y = 6$

B = (−17, 6)

***11** **P and Q are points of trisection of AB (P closer to A). How would you find P and Q using midpoints? Do it for A (4, 7) and B (16, −2).**

✔ ANSWER

AP = PQ = QB. So P is the midpoint of AQ and Q is the midpoint of PB.

Solving these gives $P = ((2x_1 + x_2)/3, (2y_1 + y_2)/3)$ and $Q = ((x_1 + 2x_2)/3, (y_1 + 2y_2)/3)$, where $A = (x_1, y_1)$, $B = (x_2, y_2)$.

$P = ((8 + 16)/3, (14 - 2)/3) = (24/3, 12/3) =$ **(8, 4)**

$Q = ((4 + 32)/3, (7 - 4)/3) = (36/3, 3/3) =$ **(12, 1)**

Check: midpoint of A (4, 7) and Q (12, 1) = (8, 4) = P ✓; midpoint of P (8, 4) and B (16, −2) = (12, 1) = Q ✓

***12 (i)** **Show that A (1, −8), B (−4, 7) and C (−7, −4) lie on a circle K with centre O (0, 0). What is the radius of K?**

✔ ANSWER

OA = $\sqrt{(1^2 + 8^2)} = \sqrt{65}$

OB = $\sqrt{(4^2 + 7^2)} = \sqrt{65}$

OC = $\sqrt{(7^2 + 4^2)} = \sqrt{65}$

All three are at the same distance from O, so they lie on a circle with centre O.

Radius = **$\sqrt{65} \approx 8.06$ units**

***12 (ii)** **Check whether D (−5, 6) and E (0, 9) lie within, on, or outside circle K.**

✔ ANSWER

OD = $\sqrt{(25 + 36)} = \sqrt{61} < \sqrt{65}$, so **D lies inside** the circle.

OE = $\sqrt{(0 + 81)} = 9 = \sqrt{81} > \sqrt{65}$, so **E lies outside** the circle.

***13** **The midpoints of the sides of triangle ABC are D (5, 1), E (6, 5) and F (0, 3). Find A, B and C.**

✔ ANSWER

Take D = midpoint of BC, E = midpoint of CA, F = midpoint of AB.

Then B + C = 2D = (10, 2), C + A = 2E = (12, 10), A + B = 2F = (0, 6) (adding x's and y's separately).

Adding all: 2(A + B + C) = (22, 18), so A + B + C = (11, 9).

A = (11, 9) − (10, 2) = **(1, 7)**

B = (11, 9) − (12, 10) = **(−1, −1)**

C = (11, 9) − (0, 6) = **(11, 3)**

Check: midpoint of BC = (5, 1) ✓, CA = (6, 5) ✓, AB = (0, 3) ✓

(If the midpoints are paired with other sides, the same three points come out with the letters swapped.)

14 (i) **Using 1 cm = 200 m, draw a model of the city (10 streets in each direction, 200 m apart). Represent the roads/streets by single lines.**

💡 **SAMPLE ANSWER**

200 m = 1 cm, so the streets are drawn 1 cm apart.

Draw 10 vertical lines (N–S streets) 1 cm apart and 10 horizontal lines (E–W streets) 1 cm apart.

Mark the two main roads (crossing at the city centre) darker; they act like the y-axis (N–S) and x-axis (E–W).

14 (ii) **Using the convention (N–S street number, E–W street number), find: (a) how many street intersections can be referred to as (4, 3); (b) how many can be referred to as (3, 4).**

✔ ANSWER

(a) The 4th N–S street and the 3rd E–W street meet at exactly one point: **only one** intersection is (4, 3).

(b) The 3rd N–S street and the 4th E–W street meet at exactly one point: **only one** intersection is (3, 4).

(4, 3) and (3, 4) are different crossings, so the order of the numbers matters, just like coordinates.

15 (i) **Screen 800 × 600 pixels, origin at the bottom-left. Circle 1: centre A (100, 150), radius 80. Circle 2: centre B (250, 230), radius 100. Does any part of either circle lie outside the screen?**

✔ ANSWER

Circle 1: x from 100 − 80 = 20 to 100 + 80 = 180; y from 150 − 80 = 70 to 150 + 80 = 230.

Circle 2: x from 250 − 100 = 150 to 250 + 100 = 350; y from 230 − 100 = 130 to 230 + 100 = 330.

All these lie between 0 and 800 (x) and 0 and 600 (y).

No part of either circle lies outside the screen.

15 (ii) **Do the two circles intersect each other?**

✔ ANSWER

AB = $\sqrt{((250 - 100)^2 + (230 - 150)^2)} = \sqrt{(150^2 + 80^2)} = \sqrt{(22500 + 6400)} = \sqrt{28900} = 170$

Sum of radii = 80 + 100 = 180; difference = 100 − 80 = 20.

$20 < 170 < 180$: the centres are closer than the sum of radii, but neither circle is inside the other.

So **yes, the circles intersect** (they cross at two points).

16 **Plot A (2, 1), B (−1, 2), C (−2, −1), D (1, −2). Is ABCD a square? Why? What is its area?**

✔ ANSWER

AB = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$

BC = $\sqrt{(1^2 + 3^2)} = \sqrt{10}$

CD = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$

DA = $\sqrt{(1^2 + 3^2)} = \sqrt{10}$

Diagonals: AC = $\sqrt{(4^2 + 2^2)} = \sqrt{20}$, BD = $\sqrt{(2^2 + 4^2)} = \sqrt{20}$

All four sides are equal and the diagonals are equal, so **ABCD is a square**.

Area = side^2 = $(\sqrt{10})^2$ = **10 square units**

🎉 **Chapter 1 done! Revise the answers once more tomorrow.**